UNIT - I

MATHEMATICAL LOGIC

PROPOSITIONAL LOGIC

Proposition :-

A declarative sentence (or assertion) which is true or false, but not both is called a proposition (or statement). Sentences which are exclamatory, interrogative or imperative in nature are not propositions.

Examples :-

1. New Delhi is the capital of India. (T) — proposition.
2. $2+2=5$ (F) — proposition.
3. How beautiful is Rose? - Not proposition.

CONNECTIVES :

Conjunction :

When p and q are any two propositions, the proposition "p and q" denoted by $p \wedge q$ and called the conjunction of p and q is defined as the compound proposition that is true when both p and q are true and is false otherwise.

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Disjunction:

When p and q are any two propositions the proposition "p or q" denoted by p∨q and called the disjunction of p and q is defined as the compound proposition that is false when both p and q are false and is true otherwise.

Negation :-

Given any proposition p, another proposition formed by writing "It is false that" before p suitably is called negation of p and is denoted by ¬p.

p	¬p
T	F
F	T

p	q	p∧q	p∨q
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Conditional and Biconditional Propositions:-

If p and q are propositions, the compound proposition "if p, then q" that denoted by p→q is called a conditional proposition, which is false when p is true and q is false and true otherwise.

If p and q are propositions, the compound proposition "p if and only if q", that is denoted by p↔q is called a

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biconditional proposition, which is true when p and q have the same truth values and is false otherwise.

P	q	$p \to q$	$p \leftrightarrow q$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

Tautology and Contradiction:-

A compound proposition $P = P(p_1, p_2, \ldots, p_n)$ where $p_1, p_2, \ldots, p_n$ are variables (elemental propositions), is called a tautology, if it true for every truth assignment for $p_1, p_2, \ldots p_n$.

P is called a contradiction, if it is false for every truth assignment for $p_1, p_2, p_3, \ldots, p_n$.

Equivalence of Propositions:-

Two compound propositions $A(p_1, p_2, \ldots, p_n)$ and $B(p_1, p_2, \ldots, p_n)$ are said to be logically equivalent or simply equivalent if they have identical truth tables, viz. if the truth value of A is equal to the truth value of B for every one of the 2^n possible sets of truth values assigned to $p_1, p_2, \ldots p_n$.

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Ex :-

$p \to q \iff \neg p \vee q$.

$\neg(p \vee q) \iff \neg p \wedge \neg q$.

p	q	$\neg p$	$\neg q$	$p \to q$	$\neg p \vee q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	T	F	F
T	F	F	T	F	F	F	F
F	T	T	F	T	T	F	F
F	F	T	T	T	T	T	T

Duality Law :-

The dual of a compound proposition that contains only the logical operators $\vee$, $\wedge$ and $\neg$ is the proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each T by F and each F by T, where T and F are special variables representing compound propositions that are tautologies and contradictions respectively. The dual of a proposition A is denoted by A^*.

Duality Theorem :-

If $A(p_1, p_2, \ldots, p_n) \equiv B(p_1, p_2, \ldots, p_n)$ where A and B are compound propositions then $A^*(p_1, p_2, \ldots, p_n) \equiv B^*(p_1, p_2, \ldots, p_n)$.

Note :-

$p \to q \equiv \neg p \vee q$.

$p \leftrightarrow q = (p \to q) \wedge (q \to p)$.

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CONVERSE, CONTRAPOSITIVE AND INVERSE STATEMENTS:

If $p \to q$ is a conditional statement, then

(i) $q \to p$ is called converse of $p \to q$.

(ii) $\neg q \to \neg p$ is called contrapositive of $p \to q$.

(iii) $\neg p \to \neg q$ is called inverse of $p \to q$.

Example,

Consider,

p: It rains.

q: The crops will grow.

$p \to q$: If it rains, then the crops will grow.

$q \to p$: If the crop grows, then there has been rain. (converse).

$\neg q \to \neg p$: If the crops do not grow then there has been no rain. (contrapositive)

$\neg p \to \neg q$: If it does not rain then the crops will grow. (Inverse).

Note:

* Conditional proposition and its converse or inverse are not logically equivalent.
* Conditional proposition and its contrapositive are logically equivalent.

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Problems:

1) Symbolize the statement, "If either Ram takes C++ or Kumar takes pascal, then Latha will take Lotus".

Soln:- Let R : Ram takes C++.
K : Kumar takes Pascal.
L : Latha takes Lotus.

∴ Symbolic form : $(R \vee K) \rightarrow L$

2. Symbolize the following statements
(i) If it is raining, then there are clouds in the sky.
(ii) If it is raining, then the sun is not shining and there are clouds in the sky.
(iii) The sun is shining if and only if it is not shining.

Soln:- Let p : It is raining
q : The sun is shining
r : There are clouds in the sky.

Symbolic form:
(i) $p \rightarrow r$
(ii) $\neg p \rightarrow (\neg q \rightarrow r)$
(iii) $q \rightarrow \neg p$.

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Laws of Algebra of Propositions:

Name of the Law	Primal form	Dual form.
1. Idempotent law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2. Identity law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3. Dominant law	$p \wedge F \equiv F$	$p \vee T \equiv T$
4. Complement law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5. Commutative law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6. Associative law	$p \vee (q \vee r) = (p \vee q) \vee r$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7. Distributive law	$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8. Absorption law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9. DeMorgan's law	$\neg (p \vee q) \equiv \neg p \wedge \neg q$	$\neg (p \wedge q) \equiv \neg p \vee \neg q$

Problems :-

1. Without using truth table prove that

$(\neg p \vee q) \wedge (p \wedge (p \wedge q)) \equiv p \wedge q$.

Soln:-

$(\neg p \vee q) \wedge (p \wedge (p \wedge q)) \equiv (\neg p \vee q) \wedge ((p \wedge p) \wedge q)$ (∵ associative law)

$\equiv (\neg p \vee q) \wedge (p \wedge q)$ (∵ Idempotent law)

$\equiv q \wedge (\neg p \vee q) \wedge p)$ (∵ associative law)

$= [(\neg p \wedge p) \vee (q \wedge p)] \wedge q$ (∵ Distributive law)

$= (F \vee (q \wedge p)) \wedge q$ (∵ complement law)

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$\equiv (q \wedge p) \wedge q$ (∵ Identity law)

$\equiv (p \wedge q) \wedge q$ (∵ commutative law)

$\equiv p \wedge (q \wedge q)$ (∵ associative law)

$\equiv p \wedge q$ (∵ Idempotent law)

2. Without truthtable prove that

$p \to (q \to p) \equiv \neg p \to (p \to q)$.

Soln:-

LHS: $p \to (q \to p)$

$\equiv \neg p \vee (\neg q \vee p)$ (∵ $p \to q \equiv \neg p \vee q$)

$\equiv \neg p \vee (p \vee \neg q)$ (∵ commutative law)

$\equiv (\neg p \vee p) \vee \neg q$ (∵ associative law)

$\equiv T \vee \neg q$ (∵ complement law)

$\equiv T$ ⟶ (1) (∵ Dominant law)

RHS: $\neg p \to (p \to q)$

$\equiv p \vee (\neg p \vee q)$ (∵ $p \to q \equiv \neg p \vee q$)

$\equiv (p \vee \neg p) \vee q$ (∵ associative law)

$\equiv T \vee q$ (∵ complement law)

$\equiv T$ ⟶ (2) (∵ Dominant law).

from (1) & (2),

LHS ≡ RHS .

$p \to (q \to p) \equiv \neg p \to (p \to q)$.

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3. Without truth table prove the following

(i) $\neg(p \leftrightarrow q) \equiv (p \vee q) \wedge \neg(p \wedge q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

(ii) $\neg((\neg p \wedge q) \vee (\neg p \wedge \neg q)) \vee (p \wedge q) \equiv p$.

Solu:-

(i) $\neg(p \leftrightarrow q) \equiv (p \vee q) \wedge \neg(p \vee q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

LHS : $\neg(p \leftrightarrow q)$

$\equiv \neg((p \rightarrow q) \wedge (q \rightarrow p))$ ($\because p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$)

$\equiv \neg[(\neg p \vee q) \wedge (\neg q \vee p)]$ ($\because p \rightarrow q \equiv \neg p \vee q$)

$\equiv (p \wedge \neg q) \vee (q \wedge \neg p)$ ($\because$ De-Morgan's law)

$\therefore \neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

$\equiv ((p \wedge \neg q) \vee \neg p) \wedge ((p \wedge \neg q) \vee q)$

($\because$ Distributive law)

$\equiv (p \vee \neg p) \wedge (\neg q \vee \neg p) \wedge (p \vee q) \wedge (\neg q \vee q)$ ($\because$ Distributive)

$\equiv T \wedge \neg(p \wedge q) \wedge (p \vee q) \wedge T$ ($\because$ De Morgan's law)

$\equiv (p \vee q) \wedge \neg(p \wedge q)$ ($\because$ Commutative law)

(ii) $\neg((\neg p \wedge q) \vee (\neg p \wedge \neg q)) \vee (p \wedge q) \equiv p$

LHS : $\neg((\neg p \wedge q) \vee (\neg p \wedge \neg q)) \vee (p \wedge q)$

$\equiv [(p \vee \neg q) \wedge (p \vee q)] \vee (p \wedge q)$ ($\because$ De-Morgan's law)

$\equiv (p \vee (\neg q \wedge q)) \vee (p \wedge q)$ ($\because$ Distributive)

$\equiv p \vee F \vee (p \wedge q)$ ($\because$ Distributive)

$\equiv p \vee (p \wedge q)$ ($\because$ Identity)

$\equiv p$ ($\because$ Absorption law)

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Tautological Implication:-

A compound proposition $A(p_1, p_2, \ldots)$ is said to tautologically imply or simply imply the compound proposition $B(p_1, p_2, \ldots, p_n)$, if B is true or equivalently if and only if $A \to B$ is a tautology. This is denoted by $A \Rightarrow B$.

Implications.

1. $p \wedge q \Rightarrow p$
2. $p \wedge q \Rightarrow q$
3. $p \Rightarrow p \vee q$
4. $\neg p \Rightarrow p \to q$
5. $q \Rightarrow p \to q$
6. $\neg(p \to q) \Rightarrow p$
7. $\neg(p \to q) \Rightarrow \neg q$
8. $p \wedge (p \to q) \Rightarrow q$
9. $\neg q \wedge (p \to q) \Rightarrow \neg p$
10. $\neg p \wedge (p \vee q) \Rightarrow q$
11. $(p \to q) \wedge (q \to r) \Rightarrow p \to r$
12. $(p \vee q) \wedge (p \to r) \wedge (q \to r) \Rightarrow r$

1. Prove the following implications without using truth tables.

   (i) $(p \vee q) \wedge (p \to r) \wedge (q \to r) \Rightarrow r$.

   (ii) $((p \vee \neg p) \to q) \to ((p \vee \neg p) \to r) \Rightarrow q \to r$.

Soln:- (i) To p.T $(p \vee q) \wedge (p \to r) \wedge (q \to r) \Rightarrow r$

It is enough to prove $[(p \vee q) \wedge (p \to r) \wedge (q \to r)] \to r$ is a tautology.

$$[(p \vee q) \wedge (p \to r) \wedge (q \to r)] \to r$$
$$\equiv [(p \vee q) \wedge ((p \vee q) \to r)] \to r$$
$$\equiv [(p \vee q) \wedge (\neg(p \vee q) \vee r)] \to r$$

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$\equiv [((p \vee r) \wedge \neg(p \vee q)) \vee ((p \vee q) \wedge r)] \rightarrow r.$

$\equiv [(p \vee r) \wedge ((p \vee q) \wedge r)] \rightarrow r.$

$\equiv ((p \vee q) \wedge r) \rightarrow r.$

$\equiv \neg((p \vee q) \wedge r) \vee r$

$\equiv \neg((p \wedge r) \vee (q \wedge r)) \vee r.$

$\equiv [(\neg p \vee \neg r) \wedge (\neg q \vee \neg r)] \vee r$

$\equiv (\neg p \vee \neg r \vee r) \wedge (\neg q \vee \neg r \vee r)$

$\equiv (\neg p \vee T) \wedge (\neg q \vee T)$

$\equiv T \wedge T$

$\equiv T$ is a tautology.

(ii) $((p \vee \neg p) \rightarrow q) \rightarrow ((p \vee \neg p) \rightarrow r) \Rightarrow q \rightarrow r$

It is enough to P.T

$[((p \vee \neg p) \rightarrow q) \rightarrow ((p \vee \neg p) \rightarrow r)] \rightarrow (q \rightarrow r)$ is a tautology.

$[((p \vee \neg p) \rightarrow q) \rightarrow ((p \vee \neg p) \rightarrow r)] \rightarrow (q \rightarrow r)$

$\equiv [(T \rightarrow q) \rightarrow (T \rightarrow r)] \rightarrow (q \rightarrow r)$

$\equiv [(F \vee q) \rightarrow (F \vee r)] \rightarrow (q \rightarrow r)$

$\equiv (q \rightarrow r) \rightarrow (q \rightarrow r)$

$\equiv \neg(q \rightarrow r) \vee (q \rightarrow r)$

$\equiv T$ is a tautology.

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3. Prove that $p \to ((q \to r)) \Rightarrow (p \to q) \to (p \to r)$.

Soln:- It is enough to P.T

$[p \to (q \to r)] \to [(p \to q) \to (p \to r)]$ is a tau

$(p \to (q \to r)) \to ((p \to q) \to (p \to r))$

$\equiv \neg(\neg p \vee (\neg q \vee r)) \vee (\neg(\neg p \vee q) \vee (\neg p \vee r))$

$\equiv (p \wedge (q \wedge \neg r)) \vee ((p \wedge \neg q) \vee (\neg p \vee r))$

$\equiv \left[(p \vee (p \wedge \neg q)) \vee ((q \wedge \neg r) \vee (p \wedge \neg q))\right] \vee (\neg p \vee r)$

$\equiv (p \wedge [(q \wedge \neg r) \vee (p \wedge \neg q)]) \vee (\neg p \vee r)$

$\equiv [p \wedge (q \wedge \neg r)] \vee ((p \wedge (p \wedge \neg q)) \vee (\neg p \vee r))$

$\equiv (p \wedge (q \wedge \neg r)) \vee ((p \wedge \neg q) \vee (\neg p \vee r))$

$\equiv [p \wedge (q \wedge \neg r)] \vee [(p \vee (\neg p \vee r)) \wedge (\neg q \vee \neg p \vee r)]$

$\equiv [p \wedge (q \wedge \neg r)] \vee [(\neg p \vee r) \wedge (\neg p \vee \neg q \vee r)]$

$\equiv (p \wedge (q \wedge \neg r)) \vee ((\neg p \vee r) \wedge (\neg p \vee \neg q \vee r))$

$\equiv (p \wedge q \wedge \neg r) \vee (\neg p \vee r \vee \neg q \vee r)$

$\equiv (p \wedge q \wedge \neg r) \vee \neg(p \wedge q \wedge \neg r)$

$\equiv T$ is a tautology.

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Q) Construct a truth table for each of the following propositions.

a) $(p \vee q) \rightarrow (p \wedge q)$

b) $(p \leftrightarrow q) \leftrightarrow ((p \wedge q) \vee (\neg p \wedge \neg q))$

c) $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$

Soln:-

(a)

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \rightarrow (p \wedge q)$
T	F	T	F	F
T	T	T	T	T
F	F	F	F	T
F	T	T	F	F

(b).

p	q	$\neg p$	$\neg q$	$p \leftrightarrow q \equiv s$	$p \wedge q \equiv a$	$\neg p \wedge \neg q \equiv b$	$a \vee b$	$s \rightarrow (a \vee b)$
T	T	F	F	T	T	F	T	T
T	F	F	T	F	F	F	F	T
F	T	T	F	F	F	F	F	T
F	F	T	T	T	F	T	T	T

(c)

p	q	r	$p \rightarrow q$	$p \rightarrow r$	$q \rightarrow r$	$p \rightarrow (q \rightarrow r) \equiv a$	$(p \rightarrow q) \rightarrow (p \rightarrow r) \equiv b$	$a \rightarrow b$.
T	T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F	T
T	F	T	F	T	T	T	T	T
T	F	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T	T
F	T	F	T	T	F	T	T	T
F	F	T	T	T	T	T	T	T
F	F	F	T	T	T	T	T	T

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Q. Determine which of the following compound propositions are tautologies and which are contradictions using truth tables.

(a) $(\neg q \wedge (p \to q)) \to \neg p$

b) $\neg(q \to r) \wedge r \wedge (p \to q)$

(a).

p	q	$\neg p$	$\neg q$	$p \to q$	$\neg q \wedge (p \to q) \equiv s$	$s \to \neg$
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	F	T
F	F	T	T	T	T	T

Since the truth value of the given compound proposition is T for all combination of p & r, it is a tautology.

(b)

p	q	r	$p \to q \equiv b$	$q \to r$	$\neg(q \to r)$	$\neg(q \to r) \wedge r \equiv a$	$a \wedge b$
T	T	T	T	T	F	F	F
T	T	F	T	F	T	F	F
T	F	T	F	T	F	F	F
T	F	F	F	T	F	F	F
F	T	T	T	T	F	F	F
F	T	F	T	F	T	F	F
F	F	T	T	T	F	F	F
F	F	F	T	T	F	F	F

The last column contains only F. It is a contradiction.

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Problem:- Find whether the conclusion C follows from the premises H_1, H_2 and H_3 in the following cases, using truth table technique

(i) $H_1 : \neg P$, $H_2 : P \vee q$; $C : P \wedge q$

solution:-

p	q	$H_1 \equiv \neg P$	$H_2 \equiv P \vee q$	$H_1 \wedge H_2$	$C \equiv P \wedge q$
T	T	F	T	F	T
T	F	F	T	F	F
F	T	T	T	T	F
F	F	T	F	F	F

H_1 and H_2 and hence $H_1 \wedge H_2$ are true in the third row in which C is false

Hence C does not follow from H_1 and H_2

(ii) $H_1 : P \vee q$, $H_2 : p \to r$; $H_3 : q \to r$ $C : r$

p	q	r	$H_1(P \vee q)$	$H_2(p \to r)$	$H_3(q \to r)$	$H_1 \wedge H_2 \wedge H_3$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	T	T	T	T
T	F	F	T	F	T	F
F	T	T	T	T	T	T
F	T	F	T	F	F	F
F	F	T	F	T	T	F
F	F	F	F	T	T	F

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H_1, H_2, H_3 and hence $H_1 \wedge H_2 \wedge H_3$ are true in the first, third and fifth rows in which r is also true.

Hence C follows from H_1, H_2 and H_3.

Problem:- Show that $(t \wedge s)$ can be derived from the premises $p \to q, q \to \neg r, r, p \vee (t \wedge s)$

Solution:-

Step NO	Statement	Reason
1	$p \to q$	P
2	$q \to \neg r$	P
3	$p \to \neg r$	T, 1, 2
4	$r \to \neg p$	T, 3
5	r	P
6	$\neg p$	T 4,5 and Modus ponens
7	$p \vee (t \wedge s)$	P
8	$t \wedge s$	T, 6, 7 and Disjunctive syllogism

Problem:- Show that $(a \vee b)$ follows logically from the premises $p \vee q, (p \vee q) \to \neg r, \neg r \to (s \wedge \neg t)$ and $(s \wedge \neg t) \to (a \vee b)$

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Step NO	Statement	Reason
1	$(P \vee q) \rightarrow \neg r$	P
2	$\neg r \rightarrow (s \wedge \neg t)$	P
3	$(P \vee q) \rightarrow (s \wedge \neg t)$	T, 1, 2 hypothetical syllogism
4	$P \vee q$	P
5	$s \wedge \neg t$	T, 3, 4 and modus ponens
6	$(s \wedge \neg t) \rightarrow (a \vee b)$	P
7	$a \vee b$	T, 5, 6 and Modus ponens

problem:- show that $(a \rightarrow b) \wedge (a \rightarrow c)$, $\neg(b \wedge c)$, $(d \vee a) \Rightarrow d$

solution:

Step NO	Statement	Reason
1	$(a \rightarrow b) \wedge (a \rightarrow c)$	P
2	$a \rightarrow b$	T, 1 and simplification
3	$a \rightarrow c$	T, 1 and simplification
4	$\neg b \rightarrow \neg a$	T, 2 and simplification
5	$\neg c \rightarrow \neg a$	T, 3 and simplification
6	$(\neg b \vee \neg c) \rightarrow \neg a$	T, 4 and 5
7	$\neg(b \wedge c) \rightarrow \neg a$	T and demorgan's law
8	$\neg(b \wedge c)$	P
9	$\neg a$	T, 7 and 8 modusponens
10	$d \vee a$	P
11	$(d \vee a) \wedge \neg a$	T, 9 and 10 conjuction
12	$(d \wedge \neg a) \vee (a \wedge \neg a)$	T, 11 and distributive law

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13	$(d \wedge \neg q) \vee F$	T, 12 and negation law
14	$d \wedge \neg q$	T, 13 and identity law
15	d	T, 14 and simplification

Problem:- give a direct proof for the implication

$p \to (q \to s)$, $(\neg r \vee p)$, $q \Rightarrow (r \to s)$.

Solution:-

Step No.	Statement	Reason
1	$\neg r \vee p$	P
2	$r \to p$	T, 1 and equivalence of
3	$p \to (q \to s)$	P
4	$r \to (q \to s)$	T, 2, 3 and hypothetical syllogism
5	$\neg r \vee (\neg q \vee s)$	T, 4 and equivalence of
6	q	P
7	$q \wedge (\neg r \vee \neg q \vee s)$	T, 5, 6 and conjunction
8	$q \wedge (\neg r \vee s)$	T, 7, 8 and negation and domination laws
9	$\neg r \vee s$	T, 8 and simplification
10	$r \to s$	T, 9 and equivalence of (9).

Problem: Use the indirect method to show that

$r \to \neg q$, $r \vee s$, $s \to \neg q$, $p \to q \Rightarrow \neg p$.

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Solution:- To use the indirect method, we will conclude $\neg\neg P \equiv P$ as an additional premise and prove by contradiction

Step No	Statement	Reason
		P (additional)
1	p	P
2	$p \to q$	T, 1, 2 and modus ponens
3	q	P
4	$r \to \neg q$	P
5	$s \to \neg q$	T, 4, 5 and equivalence
6	$(r \vee s) \to \neg q$	P
7	$r \vee s$	T, 6, 7 and modus ponens
8	$\neg q$	T, 3, 8 and conjunction
9	$q \wedge \neg q$	T, 9 and negative law
10	F	

Problem: Show that b can be derived from the premises $a \to b$, $c \to b$, $d \to (a \wedge c)$, q by indirect method.

Solution:- Let us assume $\neg b$ as an additional premise and prove by contradiction

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Step No	Statement	Reason
1	$a \to b$	P
2	$c \to b$	P
3	$(a \vee c) \to b$	T, 1, 2 and equivalence
4	$d \to (a \vee c)$	P
5	$d \to b$	T, 3, 4 and hypothetical syllogism
6	d	P
7	b	T, 5, 6 and modus ponens
8	$\neg b$	P (additional)
9	$b \wedge \neg b$	T, 7, 8 and conjunction
10	F	T, 9 and negation law

Problem:- Prove that the premises $p \to q$, $q \to r$, $s \to \neg r$ and $p \wedge s$ are inconsistent.

Solution: If we drive a contradiction by using the given premises, it means that they are inconsistent.

Step No	Statement	Reason
1	$p \to q$	P
2	$q \to r$	P
3	$p \to r$	T, 1 2 and hypothetical syllogism
4	$s \to \neg r$	P

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5	$r \rightarrow \neg s$	T, 4 and contrapositive
6	$q \rightarrow \neg s$	T, 2, 5 and hypothetical syllogism
7	$\neg q \vee \neg s$	T, 6 and equivalence (6)
8	$\neg(q \wedge s)$	T, 7 and Demorgan's law
9	$q \wedge s$	P
10	$(q \wedge s) \wedge \neg(q \wedge s)$	T, 8, 9 and conjuction
11	F	T, 10 and negation law

Problem:- Show that the following set of premises is inconsistent:

If Rama gets his degree, he will go for a job

If he goes for a job, he will get married soon.

If he goes for higher study, he will not get married.

Rama gets his degree and goes for higher study.

Solution:-

Let the statements be symbolised as:

p : Rama gets his degree

q : He will go for a job

r : He will get married soon.

s : He goes for higher study

We prove that $p \rightarrow q$, $q \rightarrow r$, $s \rightarrow \neg r$, $p \wedge s$ are inconsistent.

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Step No	Statement	Reason
1	$p \to q$	P
2	$q \to r$	P
3	$p \to r$	T, 1, 2 and hypothetical syllogism
4	$p \to s$	P
5	p	T, 4 and simplification
6	s	T, 4 and simplification
7	$s \to \neg r$	P
8	$\neg r$	T, 6, 7 and modus ponens
9	r	T, 3, 5 and modus ponens
10	$r \wedge \neg r$	T, 8, 9 and conjunction
11	F	T, 10 and negation law

Hence the set of given premises is inconsistent.

Problem:- Show that the premises "one student in this class knows how to write programs in JAVA and Everyone who knows how to write programs in JAVA can get a high-paying job" imply the conclusion "someone in this class can get a high-paying job".

Solution: C(x): x is in the class

J(x): x knows JAVA programming

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Normal Form:-

A method is to reduce $A(P_1, P_2, \ldots, P_n)$ and $B(P_1, P_2, \ldots P_n)$ to some standard forms, called normal forms and use them for deciding the nature of A or B and for comparing A and B.

There are two types of normal forms — disjunctive normal form and conjunctive normal form.

Disjunctive and Conjunctive Normal forms:-

A product of the variables and their negations (a conjunction of primary statements and their negations) is called an elementary product. For example P, $\neg P$, $P \wedge \neg P$, $P \wedge \neg q$, $q \wedge \neg P$, $P \wedge q$ are some elementary product in 2 variables.

Similarly, a sum of the variables and their negations is called an elementary sum. For examples, q, $\neg q$, $P \vee q$, $\neg q \vee \neg P$ are some elementary sum in 2 variables.

A compound proposition (or a formula) which consists of a sum of elementary products and which is equivalent to a given proposition

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is called a disjunctive normal form (DNF) of the given proposition.

A formula which consists of a product of elementary sums and which is equivalent to a given formula is called a conjunctive normal form (CNF) of the given proposition.

Max terms :-

Given a number of variables, the sum (or disjunctions) in which each variable or its negation, but not both, occurs only once are called the maxterms

Min terms :-

Given the no of variables, the products (or conjunctions) in which each variable or its negation, but not both occurs only once are called the min terms.

Note :-

1. For two variables p and q, the possible maxterms are $p \vee q$, $\neg p \vee q$, $p \vee \neg q$, $\neg p \vee \neg q$. The possible min terms are $p \wedge q$, $\neg p \wedge q$, $p \wedge \neg q$, $\neg p \wedge \neg q$.

2. For three variables p, q and r, the possible max terms are $p \vee q \vee r$, $\neg p \vee q \vee r$, $p \vee \neg q \vee r$, $p \vee q \vee \neg r$, $\neg p \vee \neg q \vee r$, $\neg p \vee q \vee \neg r$, $p \vee \neg q \vee \neg r$, $\neg p \vee \neg q \vee \neg r$.

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The possible min terms are $p \wedge q \wedge r$, $\neg p \wedge q \wedge r$, $p \wedge \neg q \wedge r$, $p \wedge q \wedge \neg r$, $\neg p \wedge \neg q \wedge r$, $\neg p \wedge q \wedge \neg r$, $p \wedge \neg q \wedge \neg r$, $\neg p \wedge \neg q \wedge \neg r$.

3. We note that there are 2^n maxterms and minterms for n variables.

PDNF and PCNF:-

A formula (compound proposition) consisting of disjunctions of minterms in the variables only and equivalent to a given formula is known as its principal disjunctive normal form (PDNF) or its sum of products canonical form of the given formula. Similarly, a formula consisting of conjunctions of maxterms in the variables only and equivalent to given formula is known as its principal conjunctive normal form (PCNF) or its product of sums canonical form.

Problems.

1. Obtain the DNF of $q \rightarrow (q \rightarrow p)$

Soln:-

$$q \rightarrow (q \rightarrow p) \equiv \neg q \vee (\neg q \vee p)$$
$$\equiv (\neg q \vee \neg q) \vee p$$
$$\equiv \neg q \vee p.$$

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2. Obtain the CNF of $\neg(p \vee q) \leftrightarrow (p \wedge q)$

Soln:-

$$\neg(p \vee q) \leftrightarrow (p \wedge q)$$

$$\equiv (\neg(p \vee q) \rightarrow (p \wedge q)) \wedge ((p \wedge q) \rightarrow \neg(p \vee q))$$

$$\equiv ((p \vee q) \vee (p \wedge q)) \wedge (\neg(p \wedge q) \vee \neg(p \vee q))$$

$$\equiv (p \vee q \vee p) \wedge (p \vee q \vee q) \wedge [(\neg p \vee \neg q) \vee (\neg p \wedge \neg q)]$$

$$\equiv (p \vee q) \wedge (p \vee q) \wedge [(\neg p \vee \neg q \vee \neg p) \wedge (\neg p \vee \neg q \vee \neg q)]$$

$$\equiv (p \vee q) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee \neg q)$$

$$\equiv (p \vee q) \wedge (\neg p \vee \neg q).$$

3. Obtain the PCNF and PDNF for the following statements.

i) $(\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q \equiv a$	$p \leftrightarrow \neg q \equiv b$	$a \rightarrow b$
T	T	F	F	F	F	T
T	F	F	T	T	T	T
F	T	T	F	T	T	T
F	F	T	T	T	F	F

PDNF of $(\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q) \equiv$

$$(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$$

Since the minterms corresponding to the 3 T values of the last column are $p \wedge q$, $p \wedge \neg q$, $\neg p \wedge q$

$\therefore$ PCNF of $(a \rightarrow b) \equiv \neg(\neg p \wedge \neg q) \equiv p \vee q$.

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(ii) $p \vee (\neg p \rightarrow (q \vee (\neg q \rightarrow r)))$

p	q	r	$\neg p$	$\neg q$	$\neg q \rightarrow r \equiv a$	$q \vee a \equiv b$	$\neg p \rightarrow b \equiv c$	$p \vee c$	Min terms	Max terms.
T	T	T	F	F	T	T	T	T	$P \wedge Q \wedge R$	
T	T	F	F	F	T	T	T	T	$P \wedge q \wedge \neg r$	
T	F	T	F	T	T	T	T	T	$P \wedge \neg q \wedge r$	
T	F	F	F	T	F	F	T	T	$P \wedge \neg q \wedge \neg r$	
F	T	T	T	F	T	T	T	T	$\neg P \wedge q \wedge r$	
F	T	F	T	F	T	T	T	T	$\neg P \wedge q \wedge \neg r$	
F	F	T	T	T	T	T	T	T	$\neg P \wedge \neg q \wedge r$	
F	F	F	T	T	F	F	F	F		$p \vee q \vee r$

$\therefore$ PDNF of the given statement

$= (P \wedge q \wedge r) \vee (P \wedge q \wedge \neg r) \vee (P \wedge \neg q \wedge r) \vee (P \wedge \neg q \wedge \neg r)$
$\vee (\neg P \wedge q \wedge r) \vee (\neg P \wedge q \wedge \neg r) \vee (\neg P \wedge \neg q \wedge r)$

PCNF of the given statement $\equiv p \vee q \vee r$.

4. Without constructing the truth table, find the principal disjunctive normal forms of the following statements, and also find the PCNF.

(i) $(\neg p \rightarrow q) \wedge (q \leftrightarrow p)$.

Soln:- Let $S \equiv (\neg p \rightarrow q) \wedge (q \leftrightarrow p) \equiv (p \vee q) \wedge ((\neg p \wedge \neg q) \vee (q \wedge p))$

$\equiv (p \vee q) \wedge ((p \wedge q) \vee \neg (p \vee q))$

$\equiv ((p \vee q) \wedge (p \wedge q)) \vee ((p \vee q) \wedge \neg (p \vee q))$

$\equiv (p \wedge (p \wedge q)) \vee (q \wedge (p \wedge q)) \vee F$

$\equiv (p \wedge q) \vee (p \wedge q)$

$\equiv p \wedge q$. which is the required PDNF.

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$\neg S \equiv$ The remaining min terms

$\equiv (\neg P \wedge q) \vee (P \wedge \neg q) \vee (\neg P \wedge \neg q)$

$\neg(\neg S) \equiv S \equiv (P \vee \neg q) \wedge (\neg P \vee q) \wedge (P \vee q)$. is the PCNF of the given statement.

5. Without constructing truth tables, find the PCNF of the following. and hence find the PDNF.

i) $(P \wedge q) \vee (\neg P \wedge q \wedge r)$.

Soln:- Let $S \equiv (P \wedge q) \vee (\neg P \wedge q \wedge r)$.

$\equiv (P \vee (\neg P \wedge q \wedge r)) \wedge (q \vee (\neg P \wedge q \wedge r))$

$\equiv (P \vee \neg P) \wedge (P \vee q) \wedge (P \vee r) \wedge (q \vee \neg P) \wedge (q \vee q) \wedge (q \vee r)$

$\equiv T \wedge (P \vee q) \wedge (P \vee r) \wedge (q \vee \neg P) \wedge q \wedge (q \vee r)$

$\equiv (P \vee q \vee F) \wedge (P \vee r \vee F) \wedge (q \vee \neg P \vee F) \wedge (q \vee F) \wedge (q \vee r \vee F)$

$\equiv (P \vee q \vee (r \wedge \neg r)) \wedge (P \vee r \vee (q \wedge \neg q)) \wedge (q \vee \neg P \vee (r \wedge \neg r)) \wedge (q \vee (P \wedge \neg P)) \wedge (q \vee r \vee (P \wedge \neg P))$

$\equiv (P \vee q \vee r) \wedge (P \vee q \vee \neg r) \wedge (P \vee r \vee q) \wedge (P \vee r \vee \neg q) \wedge (q \vee \neg P \vee r) \wedge (q \vee \neg P \vee \neg r) \wedge (q \vee P) \wedge (q \vee \neg P) \wedge (q \vee r \vee P) \wedge (q \vee r \vee \neg P)$

$\equiv (P \vee q \vee r) \wedge (P \vee q \vee \neg r) \wedge (P \vee \neg q \vee r) \wedge (P \vee \neg q \vee \neg r) \wedge (\neg P \vee q \vee r) \wedge (\neg P \vee q \vee \neg r)$

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$S \equiv \neg(\neg S) \equiv (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge \neg Q \wedge R)$

is the PDNF.

Truth Table.

P	Q	R	$\neg P$	$\neg P \to R$	$Q \leftrightarrow P$	S	Min term	Maxterm.
T	T	T	F	T	T	T	$P \wedge Q \wedge R$	
T	T	F	F	T	T	T	$P \wedge Q \wedge \neg R$	
T	F	T	F	T	F	F		$\neg P \vee Q \vee \neg R$
F	T	T	T	T	F	F		$P \vee \neg Q \vee \neg R$
T	F	F	F	T	F	F		$\neg P \vee Q \vee R$
F	T	F	T	F	F	F		$P \vee \neg Q \vee R$
F	F	T	T	T	T	T	$\neg P \wedge \neg Q \wedge R$	
F	F	F	T	F	T	F		$P \vee Q \vee R$

5. Obtain the pdnf of $\neg P \vee Q$ (or) $P \to Q$. Also find the PCNF.

Soln:-

Let $S \Leftrightarrow \neg P \vee Q$.

$\Leftrightarrow (\neg P \wedge T) \vee (Q \wedge T)$

$\Leftrightarrow (\neg P \wedge (Q \vee \neg Q)) \vee (Q \wedge (P \vee \neg P))$

$\Leftrightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge P) \vee (Q \wedge \neg P)$

$\Leftrightarrow (P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q)$

which is the P.DNF.

$\neg S \Leftrightarrow$ The remaining min terms

$\Leftrightarrow (P \wedge \neg Q)$

$S \Leftrightarrow \neg(\neg S) \Leftrightarrow \neg P \vee Q$. is the PCNF.

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$S \equiv (P \vee q \vee r) \wedge (P \vee q \vee \neg r) \wedge (P \vee \neg q \vee r)$

$\wedge (\neg P \vee q \vee r) \wedge (\neg P \vee q \vee \neg r) \wedge$

(∵ deleting repetitions)

which is the required PCNF of the given statement.

$\neg S \equiv$ The remaining maxterms

$\equiv (\neg P \vee \neg q \vee r) \wedge (P \vee \neg q \vee \neg r) \wedge (\neg P \vee \neg q \vee \neg r)$

$S \equiv \neg(\neg S) \equiv (P \wedge q \wedge \neg r) \vee (\neg P \wedge q \wedge r) \wedge (P \wedge q \wedge r)$

is the PDNF.

(ii) $(\neg P \to R) \wedge (Q \leftrightarrow P)$.

Soln:-

Let $S \equiv (\neg P \to R) \wedge (Q \leftrightarrow P)$.

$\equiv (\neg P \to R) \wedge (Q \to P) \wedge (P \to Q)$

$\equiv (P \vee R) \wedge (\neg Q \vee P) \wedge (\neg P \vee q)$

$\equiv (P \vee R \vee F) \wedge (P \vee \neg Q \vee P) \wedge (\neg P \vee Q \vee F)$

$\equiv (P \vee R \vee (Q \wedge \neg Q)) \wedge (P \vee \neg Q \vee (R \wedge \neg R))$

$\wedge (\neg P \vee Q \vee (R \wedge \neg R))$

$\equiv (P \vee Q \vee R) \wedge (P \vee \neg Q \vee R) \wedge (P \vee \neg Q \vee \neg R) \wedge (P \vee \neg Q \vee \neg R)$

$\wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$.

$\equiv (P \vee Q \vee R) \wedge (P \vee \neg Q \vee R) \wedge (\neg P \vee Q \vee R) \wedge (P \vee \neg Q \vee \neg R)$

$\wedge (\neg P \vee Q \vee \neg R)$

is the PCNF.

$\neg S \equiv$ The remaining maxterms.

$\equiv (\neg P \vee \neg Q \vee R) \wedge (P \vee Q \vee \neg R) \wedge (\neg P \vee \neg Q \vee \neg R)$

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1. Without constructing the truth tables, find the PDNF of the following statements

(i) $P \wedge \neg(q \wedge r) \vee (P \rightarrow q)$

Soln:-

$P \wedge \neg(q \wedge r) \vee (P \rightarrow q)$

$\equiv P \wedge (\neg q \vee \neg r) \vee (\neg P \vee q)$

$\equiv (P \wedge \neg q) \vee (P \wedge \neg r) \vee (\neg P \wedge (q \vee \neg q)) \vee (q \wedge (P \vee \neg P))$

$\equiv (P \wedge \neg q) \vee (P \wedge \neg r) \vee (\neg P \wedge q) \vee (\neg P \wedge \neg q) \vee (P \wedge q)$

$\equiv (P \wedge \neg q \wedge r) \vee (P \wedge \neg q \wedge \neg r) \vee (P \wedge q \wedge \neg r) \vee (P \wedge \neg q \wedge \neg r)$
$\vee (\neg P \wedge \neg q \wedge r) \vee (\neg P \wedge \neg q \wedge \neg r)$
$\vee (P \wedge q \wedge r) \vee (P \wedge q \wedge \neg r)$

$\equiv (P \wedge \neg q \wedge r) \vee (P \wedge \neg q \wedge \neg r) \vee (P \wedge q \wedge \neg r) \vee (\neg P \wedge q \wedge r)$
$\vee (\neg q \wedge \neg P \wedge r) \vee (\neg P \wedge q \wedge \neg r) \vee (\neg P \wedge \neg q \wedge \neg r)$
$\vee (P \wedge q \wedge r)$

(ii) $(q \vee (P \wedge r)) \wedge \neg((P \vee r) \wedge q)$

Soln:-

$(q \vee (P \wedge r)) \wedge \neg((P \vee r) \wedge q)$

$\equiv (q \vee (P \wedge r)) \wedge ((\neg P \wedge \neg r) \vee \neg q)$

$\equiv (q \wedge \neg P \wedge \neg r) \vee (q \wedge \neg q) \vee (P \wedge r \wedge \neg P \wedge \neg r)$
$\vee (P \vee r \wedge \neg q)$

$\equiv (\neg P \wedge q \wedge \neg r) \vee (P \wedge \neg q \wedge r)$, $(\because q \wedge \neg q \equiv F)$

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8. Without constructing the truth table, find the PCNF of the following statement

$(p \to (q \wedge r)) \wedge (\neg p \to (\neg q \wedge \neg r))$.

Soln:-

Let $S \equiv (p \to (q \wedge r)) \wedge (\neg p \to (\neg q \wedge \neg r))$

$\equiv (\neg p \vee (q \wedge r)) \wedge (p \vee (\neg q \wedge \neg r))$

$\equiv (\neg p \vee q) \wedge (\neg p \vee r) \wedge (p \vee \neg q) \wedge (p \vee \neg r)$

$\equiv (\neg p \vee q \vee (r \wedge \neg r)) \wedge ((\neg q \vee r) \vee (p \wedge \neg p))$

$\wedge (p \vee \neg q \vee (r \wedge \neg r)) \wedge (p \vee \neg r \vee (q \wedge \neg q))$

$\equiv (\neg p \vee q \vee r) \vee (\neg p \vee q \vee \neg r) \wedge (\neg q \vee r \vee p)$

$\wedge (\neg q \vee r \vee \neg p) \wedge (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee \neg r)$

$\wedge (p \vee \neg r \vee q) \wedge (p \vee \neg r \vee \neg q)$

$\equiv (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge (p \vee q \vee \neg r)$

$\wedge (\neg p \vee q \vee r) \wedge (\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg r)$

which is the PCNF.

$\neg S \equiv$ The remaining maxterms.

$\equiv (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

$S \equiv \neg(\neg S) \equiv (\neg p \wedge \neg q \wedge \neg r) \vee (p \wedge q \wedge r)$ which is the required PDNF of the given statement.

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Predicate Logic:-

A declarative sentence which can assign one of the two possible truth values true or false is called a statement.

The statements which cannot be further broken into smaller statements are called an atomic statements.

An atomic statement must contain two parts (i) subject and Declaration

Eg:- 1. Sam is a teacher
2. Ramesh is a teacher

In both the statements 'is a teacher' is the predicate and the names are called subjects.

We denote the predicates by capital letters and the subjects by small letters.

Eg:- Sam is a teacher can be represented as T(s)

There is another way to get propositions or statements from predicates which is known as quantification.

There are two types of quantifiers

1. universal quantifier
2. existential quantifier

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Eg:- $x < 5$

The statement x is less than 5 contains two parts. The first part of the variable x, is the subject of the statement. The second part is 'is less than 5' is the predicate

Propositional function at x:-

We can denote the predicate " is less than 5" and x is the variable.

$P(x)$ or P_x is called the propositional function at x.

* Once a value has been assigned to the variable x, the statement $P(x)$ becomes proposition

Eg:- The truth values of $P(2) \{\equiv 2 < 5\}$ and $P(10) \{\equiv 10 < 5\}$ are T and F respectively.

* The statement "$x = y + 5$" and "$x + y = z$" will be denoted by $P(x, y)$ or P_{xy} and $P(x, y, z)$ or P_{xyz} respectively.

Predicate Logic or Predicate calculus:-

The logic based on the analysis of predicates in any statement is called predicate logic or predicate calculus.

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Note:- An open statement P(x) is a declarative sentence which

(i) contains one or more symbols

(ii) is not a tf-statement

(iii) Produces a tf-statement when each of its symbols is replaced by a specific object from a designated set.

* The symbols appearing in open statements are called pronouns.

* The set of objects which the symbols in an open statement can represent is the universe of the open statement.

Quantifiers:-

Certain statements involve words that indicate quantity such as all, none or one. They answer the question 'How many?' since such words indicate quantity they are called quantifiers.

An analysis of mathematical sentences involving quantifiers indicates that the main two quantifiers are 'all' and 'some' where 'some' is interpreted to mean 'atleast one'.

Consider the following examples:-

1. Some men are tall

2. All birds have wings

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3. No balloon is perfectly round.

4. There is a real no. less than 10.

The quantifiers all is the universal quantifier.
We denote the symbol $\forall x$.
The quantifier some is existential quantifier
We denote the symbol $\exists x$

Write the following sentences in symbolic form:-

1. Something is Green	$\exists x\,(G(x))$
2. Everything is Green	$\forall x\,(G(x))$
3. Nothing is Green	$\forall x\,(\neg G(x))$
4. Something is not green	$\exists x\,(\neg G(x))$

We observe that

$G(x)$: x is green is an open statement and hence it is not tf-statement, but the statements obtained from it by prefixing quantifiers are tf-statements. The statements are

$(\exists x)(G(x))$ and $(\forall x)(G(x))$ are tf-statements

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$H(x)$: x can get a high paying job

Then the given premises are $\exists x\,(C(x) \wedge J(x))$

and $\forall x\,(J(x) \rightarrow H(x))$

The conclusion is $\exists x\,(C(x) \wedge H(x))$

Step No	Statement	Reason
1	$\exists x\,(C(x) \wedge J(x))$	P
2	$C(a) \wedge J(a)$	ES and 1
3	$C(a)$	T, 2 and simplification
4	$J(a)$	T, 2 and simplification
5	$\forall x\,(J(x) \rightarrow H(x))$	P
6	$J(a) \rightarrow H(a)$	US and 5
7	$H(a)$	T, 4, 6 and modusponens
8	$C(a) \wedge H(a)$	T, 3, 7 and conjunction
9	$\exists x\,(C(x) \wedge H(x))$	EG and 8

Problem:- If $L(x, y)$ symbolises the statement "x loves y", where the universe of discourse for both x and y consists of all people in the world, translate the following english sentences into logical expressions:

(a) Every body loves y.

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(b) Everybody loves somebody
(a) There is somebody whom everybody loves.
(d) Nobody loves everybody
(e) There is somebody whom no one loves.

Solution:

(a) $L(x, y)$ for all x. Hence $\forall x \; L(x, y)$

(b) $L(x, y)$ is true for all x and some y.

Hence $\forall \exists y \; L(x, y)$

(c) Even though, (e) is the same as (b), the existence of somebody (y) whom all x love

Hence $\exists y \; \forall x \; L(x, y)$

(d) No body loves everybody

(e) There is not one who loves everybody

Hence $\neg \exists x \; \forall y \; L(x, y)$

$\equiv \forall x \; \neg \forall y \; L(x, y)$

$\equiv \forall x \; \exists y \; \neg L(x, y)$

(e) that there is somebody whom every one does not love.

Hence $\neg \forall x \; \exists y \; L(x, y)$

$\equiv \exists x \; \neg \exists y \; L(x, y)$

$\equiv \exists x \; \forall y \; \neg L(x, y)$

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Quantifiers

There are two types of Quantifiers

1. Universal Quantifier
2. Existential Quantifier

Many mathematical statements, assert that a property is true for all values of a variable in a particular domain is called the universe of discourse.

Such a statement is expressed using a universal quantifier.

Universal Quantifier :-

The universal quantification of $P(x)$ is the statement.

"$P(x)$ is true for all values of x in the universe of discourse". and is denoted by the notation $(x)P(x)$ or $\forall x P(x)$.

The symbol $\forall$ is called the universal Quantifier.

It represents for all x, all x such that
for every x, every x such that
for each x, each x such that

Existential Quantifier :-

The existential quantification of $P(x)$ is the proposition

"There exists atleast one x (or an x) such that $P(x)$ is true" and is denoted by the notation $\exists x P(x)$

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Negation of Quantified statements :-

Consider the propositional function $P(x)$.

"all intergers greater than 5"

The universal quantification $\forall x P(x)$.

The negation is

"there is an integer less than or equal to 5".

(ie) $\exists$ an integer x, $x \leq 5$

Free and bound variables :-

When a quantifier is used on a predicate variable x or when a value is assigned to this variable, we say that the occurrence of the variable is bound.

An occurrence of the variable x that is not bound by a quantifier is said to be free.

Scope :-

The part of the logical expression or predicate formula to which a quantifier is applied is called the scope of the quantifier.

Predicate formula	Bound Variable & scope	Free variable
1. $\forall x\ P(x,y)$	x; $P(x,y)$	y
2. $\forall x(P(x) \rightarrow Q(x))$	x; $P(x) \rightarrow Q(x)$	—
3. $\forall x(P(x) - E(y) Q(x,y)$	x; $P(x) \rightarrow E(y) Q(x,y)$	—
4. $\forall x(P(x) \wedge Q(x)) \vee \forall y R(y)$	x; $P(x) \wedge Q(x)$ y; $R(y)$	—

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Nested Quantifiers -

Nested Quantifiers are quantifiers that occur within the scope of other quantifiers.

Eg:- $\forall x \exists y (x+y=0)$

(or) for every real number x there exists a real number y such that $x+y=0$

Inference theory of predicate calculus

Rules of Inference for quantified statements:-

* Rules of Inference
  - → Rule P
  - → Rule T
  - → Rule CP
* Rules on Quantifiers
  - → Rule US } → To eliminate quantifiers
  - → Rule ES } during the course of derivation
  - → Rule UG } - To quantify the
  - → Rule EG } desired conclusion.

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Rule US :- Universal specification is the rule of inference which states that one can conclude that P(c) is true, if $\forall x\, P(x)$ is true, where c is an arbitrary member of the universe of discourse. This rule is also called the universal instantiation.

Rule ES :- Existential specification is the rule which allows us to conclude that P(c) is true, if $\exists x\, P(x)$ is true, where c is not an arbitrary member of the universe, but one of for which P(c) is true. This rule is also called the existential instantiation.

Rule UG :- Universal Generalisation is the rule which states that $\forall x\, P(x)$ is true, if P(c) is true, where c is an arbitrary member of the universe of discourse.

Rule EG :- Existential Generalisation is the rule that is used to conclude that $\exists x\, P(x)$ is true when P(c) is true, where c is particular member of the universe of discourse.

Eg: Let us consider the following:
'Famous Socrates argument' which is

by All men are moral
Socrates is a man
therefore Socrates is a moral

Notations:- $H(x)$: x is a man
$M(x)$: x is a mortal
S : Socrates

$\forall x (H(x) \rightarrow M(x)) \wedge H(S) \Rightarrow M(S)$

Steps	Statement	Reason
1	$\forall x (H(x) \rightarrow M(x))$	P
	$H(S) \rightarrow M(S)$	US, 2
2	$H(S)$	P
3	$M(S)$	T, 2, 3 Modus ponens
4		

Example:- Show that the premises "one student in this class knows how to write programs in JAVA" and "Everyone who knows how to write programs in JAVA can get a high-paying job" imply the conclusion "someone in this class can get a high-paying job".

Sol $C(x)$: x is in this class
$J(x)$: x knows JAVA programming
$H(x)$: x can get a high paying job

Conclusion: $\exists x (C(x) \wedge J(x))$ and
$\forall x (J(x) \rightarrow H(x))$ is
$\exists x (C(x) \wedge H(x))$

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Step No.	Statement	Reason
1	$\exists x (C(x) \wedge J(x))$	P
2	$C(a) \wedge J(a)$	Es and 1
3	$C(a)$	T, 2
4	$J(a)$	T, 2
5	$\forall x (J(x) \rightarrow H(x))$	P
6	$J(a) \rightarrow H(a)$	US and 5
7	$H(a)$	T, 4, 6 and modus ponens
8	$C(a) \wedge H(a)$	T, 3, 7 and conjunction
9	$\exists x (C(x) \wedge H(x))$	EG and 8

Example:- Prove that $(\forall x)(P(x) \rightarrow \neg Q(x))$ follows from the premises

$(\exists x)(P(x) \wedge Q(x)) \rightarrow (\forall y)(R(y) \rightarrow S(y))$; $(\exists y)(R(y)) \wedge \neg S(y))$

Solution:-

Step No	Statement	Reason
1	$(\exists y)(R(y) \wedge \neg S(y))$	Rule P
2	$R(a) \wedge \neg S(a)$	Rule ES (1)
3	$T(R(a) \rightarrow S(a))$	Rule T, (2)
4	$(\exists y)(\neg(R(y) \rightarrow S(y))$	Rule EG (3)
5	$\neg(\forall y)[R(y) \rightarrow S(y)]$	Rule T (4)
6	$(\exists x)[P(x) \wedge Q(x)] \rightarrow (\forall y)[R(y) \rightarrow S(y)]$	Rule P
7	$\neg(\forall y)[R(y) \rightarrow S(y)] \rightarrow \neg(\exists x][P(x) \wedge Q(x)]$	Rule T

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8	$\neg(\exists x)(P(x) \wedge Q(x))$	Rule T (5 & 7)
9	$(\forall x) \neg(P(x) \wedge Q(x))$	Rule T
10	$\neg(P(b) \wedge Q(b))$	Rule US (9)
11	$\neg P(b) \vee \neg Q(b)$	Rule T (9)
12	$P(b) \rightarrow \neg Q(b)$	Rule T (10)
13	$(\forall x)(P(x) \rightarrow \neg Q(x))$	Rule UG (12)

Problem:- Show that $(\exists x)\, M(x)$ follows logically from the premises $(\forall x)(H(x) \rightarrow M(x))$ and $(\exists x)\, H(x)$

Solution:-

Step No	Statement	Reason
1	$(\forall x)(H(x) \rightarrow M(x))$	Rule P
2	$H(a) \rightarrow M(a)$	Rule US (1)
3	$(\exists x)\, H(x)$	Rule P
4	$H(a)$	Rule ES (3)
5	$M(a)$	Rule T (4, 5)
6	$(\exists x)\, M(x)$	Rule EG (5)

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Problem: Prove that $\forall x (P(x) \rightarrow (Q(y) \wedge R(x)))$, $(\exists x) P(x) \Rightarrow Q(y) \wedge \exists x (P(x) \wedge R(x))$

Solution:-

Step No	Statement	Reason
1	$\forall x (P(x) \rightarrow (Q(y)) \wedge R(x))$	Rule P
2	$P(y) \rightarrow (Q(y) \wedge R(y))$	Rule US (1)
3	$(\exists x) P(x)$	Rule P
4	$P(y)$	Rule ES(3)
5	$Q(y) \wedge R(y)$	Rule T (2, 4)
6	$Q(y)$	Rule T (5)
7	$R(y)$	Rule T (5)
8	$P(y) \wedge R(y)$	Rule T (4, 7)
9	$(\exists x) (P(x) \wedge R(x))$	Rule EG (8)
10	$Q(y) \wedge \exists x (P(x) \wedge R(x))$	Rule T (6, 9)

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Soundness

The statement that every provable formula is valid is known as soundness.

If A is a formula, the notation $\vdash A$ is used to express the fact that A is provable and the notation $\models A$ to express that A is valid.

In symbolic, for propositional formula A, if $\vdash A$, then $\models A$.

Completeness

Every valid formula is provable is known as Completeness.

In symbolic terms, every formula A, if $\models A$, then $\vdash A$.

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Provable:

A propositional formula is said to be provable if there is a formal proof of it in that system.

Valid

A propositional formula is said to be a tautology or valid if it is true under any truth assignment.

Syntatic

Provability is a syntactic notion which asserts the existence of a syntactic object namely a proof.

Semantic

Validity is a semantic notion which has to do with truth assignments and valuation.

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Semantics

Semantics is Concerned with the meaning of expression when the symbols are interpreted in a certain way.

The Connectives negation, Conjunction, disjunction, implication and equivalence have their intended meanings not, and, or, if then and iff.

Syntax

Syntax is Concerned with the formal structures of expressions irrespective of any interpretations.

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Satisfiability :-

Σ is satisfiable iff there is some truth valuation t such that $\Sigma^t = 1$.
When $\Sigma^t = 1$, t is said to satisfy Σ.

Obviously, if Σ is satisfiable then it implies that all the formulas in it will be satisfiable. But the converse of this implication does not hold, because the satisfiability of Σ requires a single truth valuation satisfying all the formulas in it.

Propositional Language ($\mathcal{L}^P$)

A formal language is a collection of symbols, which should be distinguished from symbols of the metalanguage used in studying them.
Symbols used are differentiated into 3 classes.

Tautological Consequence :-

Suppose $\Sigma \subseteq$ form ($\mathcal{L}^P$) and $A \in$ form ($\mathcal{L}^P$).

A is a tautological consequence of Σ written as $\Sigma \models A$, iff for any truth valuation t, $\Sigma^t = 1 \Rightarrow A^t = 1$.

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Adequate set of connectives :-

A set of connectives is said to be adequate if any n-ary ($n \geq 1$) connective can be defined interms of them.

Note :-

→ $\{\neg, \wedge, \vee\}$ is an adequate set of connectives.

→ $\{\neg, \vee\}$, $\{\neg, \wedge\}$ and $\{\neg, \rightarrow\}$ are adequate.

→ $\{\vee, \wedge\}$ is not adequate.

Formulas $A \rightarrow B$ and $\neg A \vee B$ are tautologi

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Truth Valuation

A truth valuation is a function with the set of all proposition symbols as domain and $\{1,0\}$ as range. The letter t denotes any truth valuation. The value t assigns to any formula A is written as A^t.

The value assigned to formula by a truth valuation t is defined by

(i) $p^t \in \{1,0\}$

(ii) $(\neg A)^t = \begin{cases} 1 & \text{if } A^t = 0 \\ 0 & \text{otherwise} \end{cases}$

(iii) $(A \wedge B)^t = \begin{cases} 1 & \text{if } A^t = B^t = 1 \\ 0 & \text{otherwise} \end{cases}$

(iv) $(A \vee B)^t = \begin{cases} 1 & \text{if } A^t = 1 \text{ or } B^t = 1 \\ 0 & \text{otherwise} \end{cases}$

(v) $(A \to B)^t = \begin{cases} 1 & \text{if } A^t = 0 \text{ or } B^t = 1 \\ 0 & \text{otherwise} \end{cases}$

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For -

(vi) $(A \leftrightarrow B)^t = \begin{cases} 1 & \text{if } A^t = B^t \\ 0 & \text{otherwise} \end{cases}$

The symbol Σ denotes any set of formulas.

$$\Sigma^t = \begin{cases} 1 & \text{if each } B \in \Sigma,\ B^t = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Tautological Consequence

A is a tautological consequence of Σ written as $\Sigma \models A$ iff for any truth valuation t, $\Sigma^t = 1$ implies $A^t = 1$.

$\models$ is not a symbol of the formal language and hence $\Sigma \models A$ is not a formula. $\Sigma \models A$ is a proposition about Σ and A. $\models$ may be read as "logically implies".

For two formulas $A \dashv\vdash B$ to denote "$A \models B$ and $B \models A$". A and B are said to be tautologically equivalent iff $A \dashv\vdash B$ holds.

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